Note

- A relation may be represented algebraically either by the roster method or by the set builder method.
- An arrow diagram is a visual representation of a relation.

Example 1.5 The arrow diagram shows (Fig.1.10) a relationship between the sets P and Q. Write the relation in (i) Set builder form (ii) Roster form (iii) What is the domain and range of R.

Solution

(i) Set builder form of R $= \{(x, y) \mid y = x - 2,\ x \in P, y \in Q\}$

(ii) Roster form R $= \{(5,3),(6,4),(7,5)\}$

(iii) Domain of R = {5,6,7} and range of R $= \{3,4,5\}$

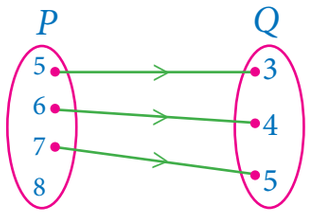


Fig. 1.10

'Null relation'

Let us consider the following example. Suppose $A = \{-3,-2,-1\}$ and $B = \{1,2,3,4\}$. A relation from A to B is defined as $a - b = 8$ i.e., there is no pair(a,b) such that $a - b = 8$. Thus R contain no element and so $R = \phi$.

A relation which contains no element is called a "Null relation".

DO YOU KNOW?

If $n(A) = p$, $n(B) = q$, then the total number of relations that exist from A to B is 2^{pq}.

Exercise 1.2

1. Let $A = \{1,2,3,7\}$ and $B = \{3,0,-1,7\}$, which of the following are relation from A to B?
   (i) $R_1 = \{(2,1), (7,1)\}$ (ii) $R_2 = \{(-1,1)\}$
   (iii) $R_3 = \{(2,-1), (7,7), (1,3)\}$ (iv) $R_4 = \{(7,-1), (0,3), (3,3), (0,7)\}$
2. Let $A = \{1,2,3,4,\ldots,45\}$ and R be the relation defined as "square is of a number" on A. Write R as a subset of $A \times A$. Also, find the domain and range of R.
3. A Relation R is given by the set $\{(x, y) / y = x + 3,\ x \in \{0,1,2,3,4,5\}\}$. Determine its domain and range.
4. Represent each of the given relations by (a) an arrow diagram, (b) a graph and (c) a set in roster form, wherever possible.
   (i) $\{(x,y) \mid x = 2y,\ x \in \{2,3,4,5\},\ y \in \{1,2,3,4\}\}$
   (ii) $\{(x,y) \mid y = x+3,\ x, y$ are natural numbers $< 10\}$
5. A company has four categories of employees given by Assistants (A), Clerks (C), Managers (M) and an Executive Officer (E). The company provide ₹10,000, ₹25,000, ₹50,000 and ₹1,00,000 as salaries to the people who work in the categories

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